Session 3.1: Foundations of Artificial Intelligence: Enterprise Value Creation and Decision Automation

TODAY'S PURPOSE

Why This Session Matters

- Understand what Artificial Intelligence (AI) can do for a business
- Recognize where it falls short
- Decide which AI projects are worth funding

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Distinguish AI, Machine Learning (ML), and Deep Learning (DL)
2. Calculate project Return on Investment (ROI)
3. Allocate a fixed budget and fix a ZeroDivisionError

Concepts for This Session

CONCEPT 1

Artificial Intelligence (AI)

Broad computational systems capable of performing tasks that typically require human cognition, including perception, language understanding, and automated decision-making.

CONCEPT 2

Machine Learning (ML)

A subfield of artificial intelligence where statistical models learn patterns and relationships directly from data rather than following rigid, hand-coded rules.

CONCEPT 3

Deep Learning (DL)

A subset of machine learning based on multi-layered artificial neural networks, specialized in unstructured data processing (images, audio, natural language).

CONCEPT 4

Rule-Based vs. Statistical Systems

Transitioning from deterministic if-then heuristics to probabilistic algorithms that generalize across noisy, unseen real-world data.

How the Concept Works in Practice

FOUNDATION 1

The Artificial Intelligence Hierarchy

Deconstructing industry hype. Machine Learning is not magic; it is applied computational statistics. AI encompasses the broader vision, ML provides data-driven statistical learning, and Deep Learning handles high-dimensional unstructured tasks.

FOUNDATION 2

The Economics of Machine Learning (Prediction as a Commodity)

When the cost of quantitative prediction drops toward zero, prediction is incorporated into every operational decision: fraud screening, demand forecasting, inventory routing, and pricing.

FOUNDATION 3

Enterprise ROI Evaluation Matrix

Evaluating AI projects requires rigorous financial modeling. Students quantify Implementation Cost (compute, labeling, salaries), Annual Labor Savings, and Projected Payback Periods.

FOUNDATION 4

Identifying High-Impact Machine Learning Use Cases

Knowing when NOT to use ML. If a business policy is strictly deterministic (such as calculating sales tax), use standard rule-based code. When relationships are complex, noisy, and dynamic (such as customer churn or fraud detection), use machine learning.

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: Foundations of Artificial Intelligence: Enterprise Value Creation and Decision Automation

```
import pandas as pd

    df = pd.read_csv("data/session_03_01_dataset.csv")

    df["Net_Annual_Savings_USD"] = df["Projected_Cost_Savings_USD"] -
(df["Implementation_Cost_USD"] * 0.15) # 15% maintenance
    df["Calculated_Payback_Months"] = (df["Implementation_Cost_USD"] /
df["Net_Annual_Savings_USD"]) * 12.0
    df["3Year_Net_ROI_Pct"] = (((df["Net_Annual_Savings_USD"] * 3.0) -
df["Implementation_Cost_USD"]) / df["Implementation_Cost_USD"]) *
100.0

    print(df[["Project_ID", "Department_Name",
"Calculated_Payback_Months", "3Year_Net_ROI_Pct"]].head(3))
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: Foundations of Artificial Intelligence: Enterprise Value Creation and Decision Automation

```
# Rule-Based Heuristic
    def rule_based_flag(spend):
        return "Flagged" if spend > 250000.0 else "Approved"

    # Probabilistic Scoring Equivalence
    def statistical_score(spend, risk_score):
        prob = (spend / 500000.0) * 0.5 + (risk_score / 100.0) * 0.5
        return "High_Risk" if prob >= 0.65 else "Low_Risk"

    print("Rule-Based Evaluation:", rule_based_flag(260000.0))
    print("Probabilistic Evaluation:", statistical_score(260000.0,
75))
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 3: Foundations of Artificial Intelligence: Enterprise Value Creation and Decision Automation

```
high_priority = df[(df["Calculated_Payback_Months"] <= 6.0) &
(df["Annual_Labor_Hours_Saved_Count"] >= 4000)]
    print(f"High Priority AI Projects Count: {len(high_priority)}")
    print(high_priority[["Project_ID", "Department_Name",
"Calculated_Payback_Months"]])
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.